

# Soil vs Built-up After Adding 1,000 MP Soil Points

Eight Madhya Pradesh districts, scored against ESA WorldCover v200

| district | best model    | overall | soil recall | Soil->Built |
|----------|---------------|---------|-------------|-------------|
| Indore   | Random Forest | 60.7%   | 19% -> 41%  | 107 -> 79   |
| Dhar     | CART          | 64.5%   | 73% -> 79%  | 32 -> 21    |
| Sagar    | Random Forest | 72.7%   | 45% -> 65%  | 64 -> 42    |
| Shajapur | CART          | 63.2%   | 64% -> 75%  | 37 -> 24    |
| Gwalior  | Random Forest | 80.2%   | 42% -> 55%  | 77 -> 60    |
| Vidisha  | Random Forest | 68.8%   | 32% -> 48%  | 92 -> 71    |
| Rewa     | CART          | 75.2%   | 67% -> 73%  | 37 -> 27    |
| Betul    | CART          | 64.5%   | 55% -> 68%  | 30 -> 19    |
| MEAN     |               |         | 50% -> 63%  | 60 -> 43    |

Mean soil recall rose 50% to 63%.

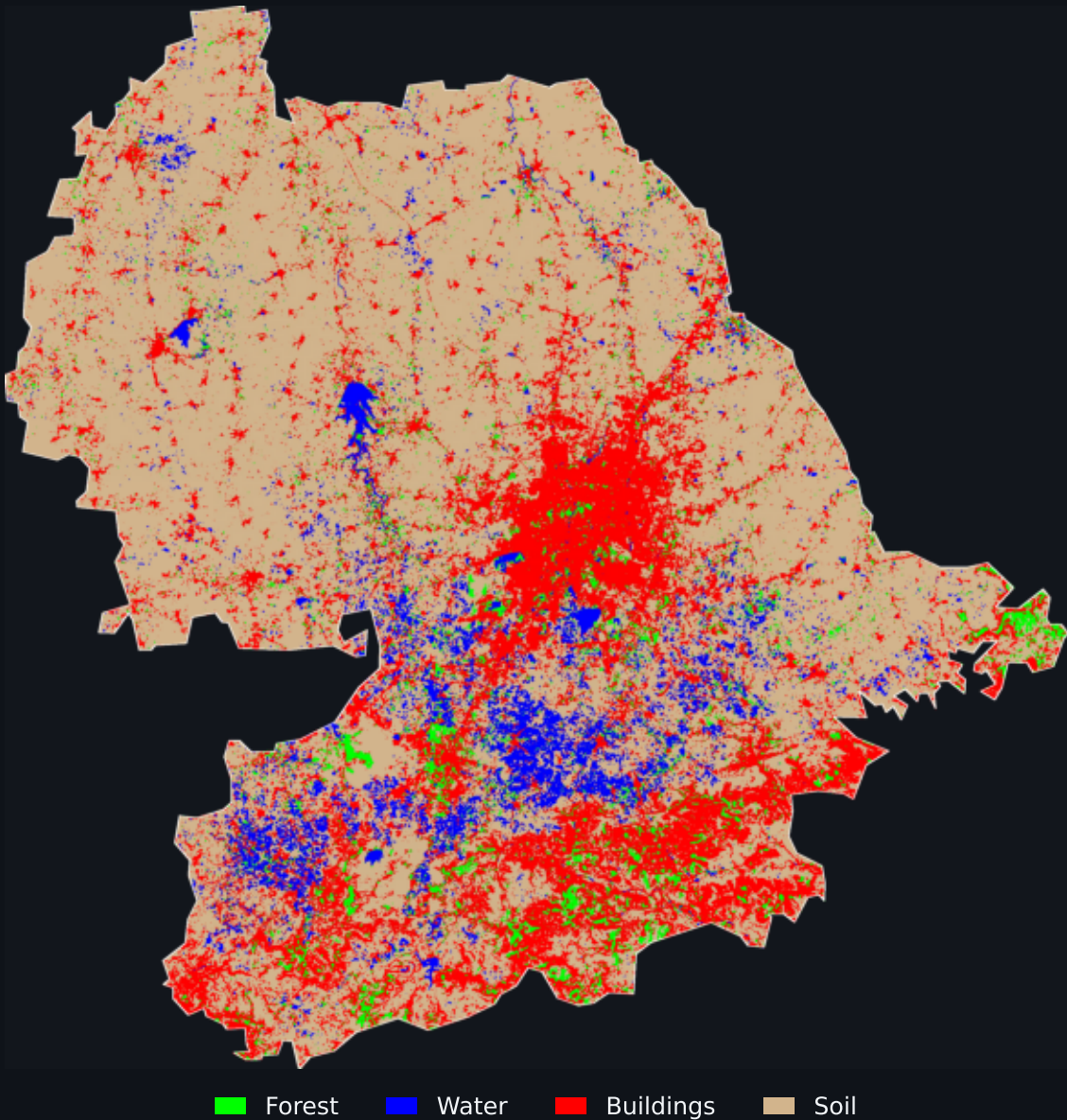
Bare ground misread as built-up fell 60 to 43 points per district,  
a 28% reduction in the project's single largest error.

Every district improved on soil recall. No district regressed on it.

Reference: ESA WorldCover v200 at 10 m, sampled at 20 m, 150 stratified points per class per district. Cropland (WorldCover class 40) is excluded: in a March-April composite a field may be bare or green, so it has no honest mapping onto four classes. WorldCover is itself only 75-80% accurate, so these are agreement figures, not ground truth.

# Indore

Best model: Random Forest · 60.7% agreement with WorldCover



## CONFIGURATION

|           |                                               |
|-----------|-----------------------------------------------|
| model     | Random Forest                                 |
| params    | 200 trees · bag 0.3 · maxNodes 10 · minLeaf 1 |
| composite | 2025-03-01 to 2025-04-30, cloud <15%          |
| features  | 27 bands (S2 + indices + GLCM + SAR)          |
| training  | 5,000 pts (4,000 Jabalpur + 1,000 MP soil)    |
| smoothing | 1 px focal mode                               |

## AGREEMENT WITH ESA WORLDCOVER

|           |        |           |
|-----------|--------|-----------|
| class     | recall | precision |
| Forest    | 29%    | 92%       |
| Water     | 81%    | 92%       |
| Buildings | 91%    | 50%       |
| Soil      | 41%    | 42%       |
| OVERALL   | 60.7%  |           |

## EFFECT OF THE 1,000 NEW SOIL POINTS

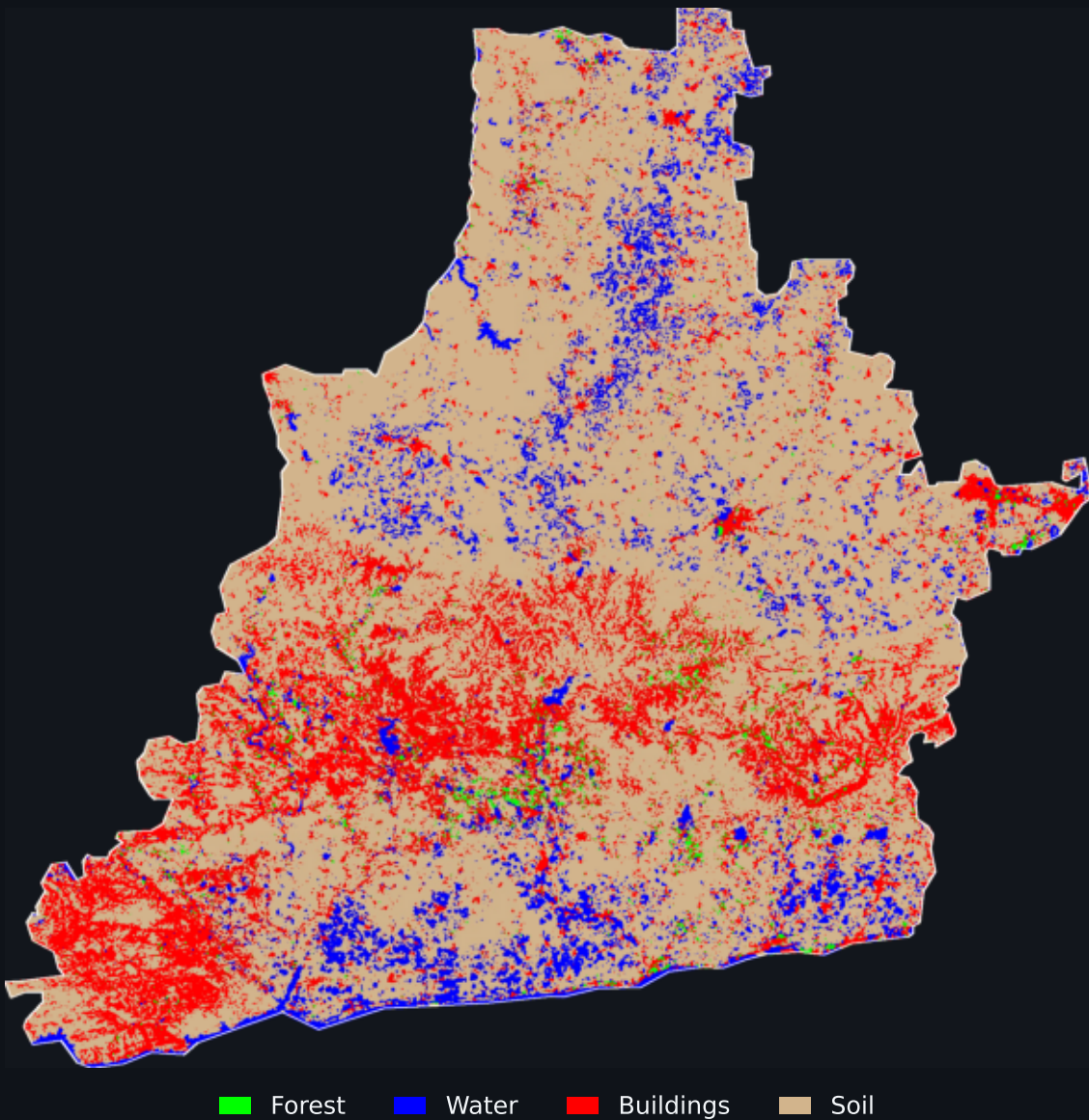
|                 |        |       |         |
|-----------------|--------|-------|---------|
|                 | before | after | change  |
| Soil recall     | 19%    | 41%   | +22.0pp |
| Buildings prec. | 43%    | 50%   | +6.7pp  |
| Soil->Buildings | 107    | 79    | -28     |
| Overall         | 57%    | 61%   | +3.5pp  |

## AREA REPORTED BY THE DASHBOARD

|           |       |            |
|-----------|-------|------------|
| Forest    | 9.0%  | 38,386 ha  |
| Water     | 3.2%  | 13,538 ha  |
| Buildings | 14.9% | 63,562 ha  |
| Soil      | 72.9% | 310,141 ha |

# Dhar

Best model: CART · 64.5% agreement with WorldCover



## CONFIGURATION

|           |                                            |
|-----------|--------------------------------------------|
| model     | CART                                       |
| params    | maxNodes 20                                |
| composite | 2025-03-01 to 2025-04-30, cloud <15%       |
| features  | 27 bands (S2 + indices + GLCM + SAR)       |
| training  | 5,000 pts (4,000 Jabalpur + 1,000 MP soil) |
| smoothing | 1 px focal mode                            |

## AGREEMENT WITH ESA WORLDCOVER

|           |        |           |
|-----------|--------|-----------|
| class     | recall | precision |
| Forest    | 17%    | 93%       |
| Water     | 81%    | 82%       |
| Buildings | 81%    | 51%       |
| Soil      | 79%    | 62%       |
| OVERALL   | 64.5%  |           |

## EFFECT OF THE 1,000 NEW SOIL POINTS

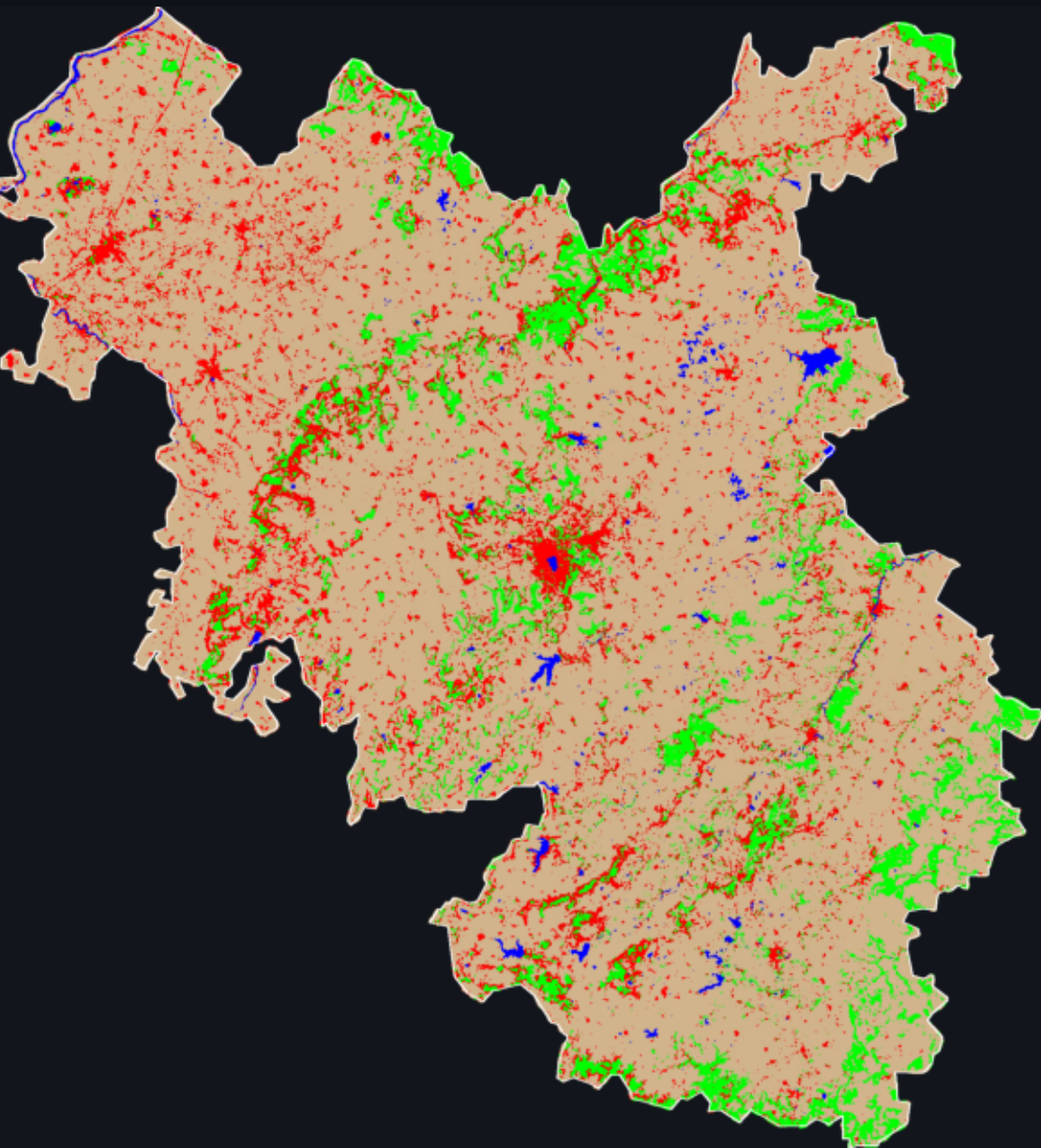
|                 |        |       |        |
|-----------------|--------|-------|--------|
|                 | before | after | change |
| Soil recall     | 73%    | 79%   | +6.0pp |
| Buildings prec. | 52%    | 51%   | -0.3pp |
| Soil->Buildings | 32     | 21    | -11    |
| Overall         | 67%    | 64%   | -2.2pp |

## AREA REPORTED BY THE DASHBOARD

|           |       |            |
|-----------|-------|------------|
| Forest    | 5.2%  | 46,363 ha  |
| Water     | 3.8%  | 33,813 ha  |
| Buildings | 17.9% | 158,311 ha |
| Soil      | 73.0% | 646,142 ha |

# Sagar

Best model: Random Forest · 72.7% agreement with WorldCover



Forest Water Buildings Soil

n = 600 stratified WorldCover points · before = 4,000 Jabalpur points, after = 5,000 including the new MP soil set  
Area breakdown omitted: the 10 m area histogram over this district exceeds Earth Engine's interactive limit.

## CONFIGURATION

|           |                                               |
|-----------|-----------------------------------------------|
| model     | Random Forest                                 |
| params    | 200 trees · bag 0.3 · maxNodes 10 · minLeaf 1 |
| composite | 2025-03-01 to 2025-04-30, cloud <15%          |
| features  | 27 bands (S2 + indices + GLCM + SAR)          |
| training  | 5,000 pts (4,000 Jabalpur + 1,000 MP soil)    |
| smoothing | 1 px focal mode                               |

## AGREEMENT WITH ESA WORLDCOVER

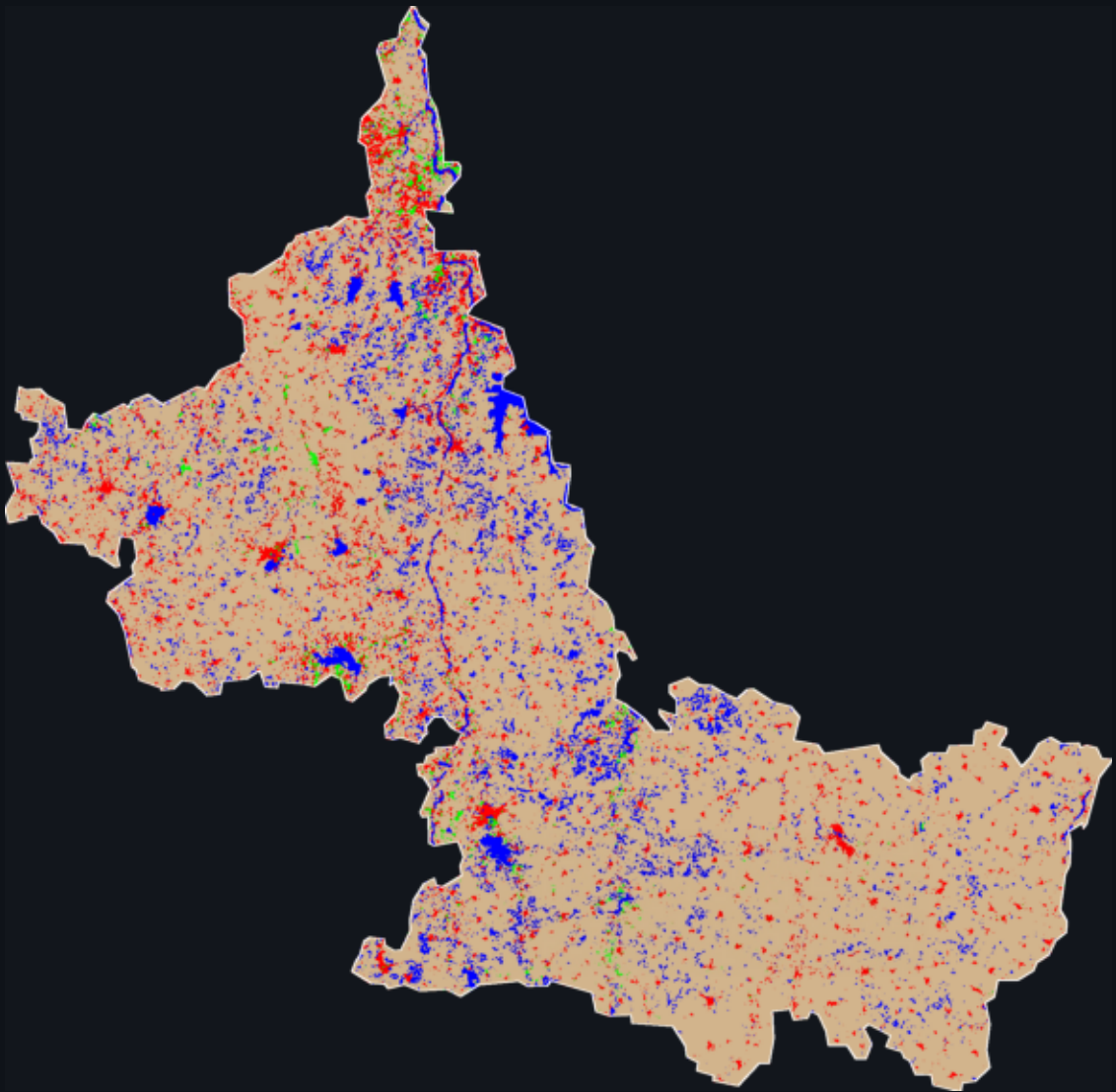
|           |        |           |
|-----------|--------|-----------|
| class     | recall | precision |
| Forest    | 59%    | 100%      |
| Water     | 76%    | 89%       |
| Buildings | 90%    | 66%       |
| Soil      | 65%    | 55%       |
| OVERALL   | 72.7%  |           |

## EFFECT OF THE 1,000 NEW SOIL POINTS

|                 |        |       |         |
|-----------------|--------|-------|---------|
|                 | before | after | change  |
| Soil recall     | 45%    | 65%   | +20.0pp |
| Buildings prec. | 58%    | 66%   | +8.3pp  |
| Soil->Buildings | 64     | 42    | -22     |
| Overall         | 72%    | 73%   | +0.8pp  |

# Shajapur

Best model: CART · 63.2% agreement with WorldCover



Forest Water Buildings Soil

## CONFIGURATION

|           |                                            |
|-----------|--------------------------------------------|
| model     | CART                                       |
| params    | maxNodes 20                                |
| composite | 2025-03-01 to 2025-04-30, cloud <15%       |
| features  | 27 bands (S2 + indices + GLCM + SAR)       |
| training  | 5,000 pts (4,000 Jabalpur + 1,000 MP soil) |
| smoothing | 1 px focal mode                            |

## AGREEMENT WITH ESA WORLDCOVER

|           |        |           |
|-----------|--------|-----------|
| class     | recall | precision |
| Forest    | 25%    | 93%       |
| Water     | 71%    | 75%       |
| Buildings | 81%    | 65%       |
| Soil      | 75%    | 49%       |
| OVERALL   | 63.2%  |           |

## EFFECT OF THE 1,000 NEW SOIL POINTS

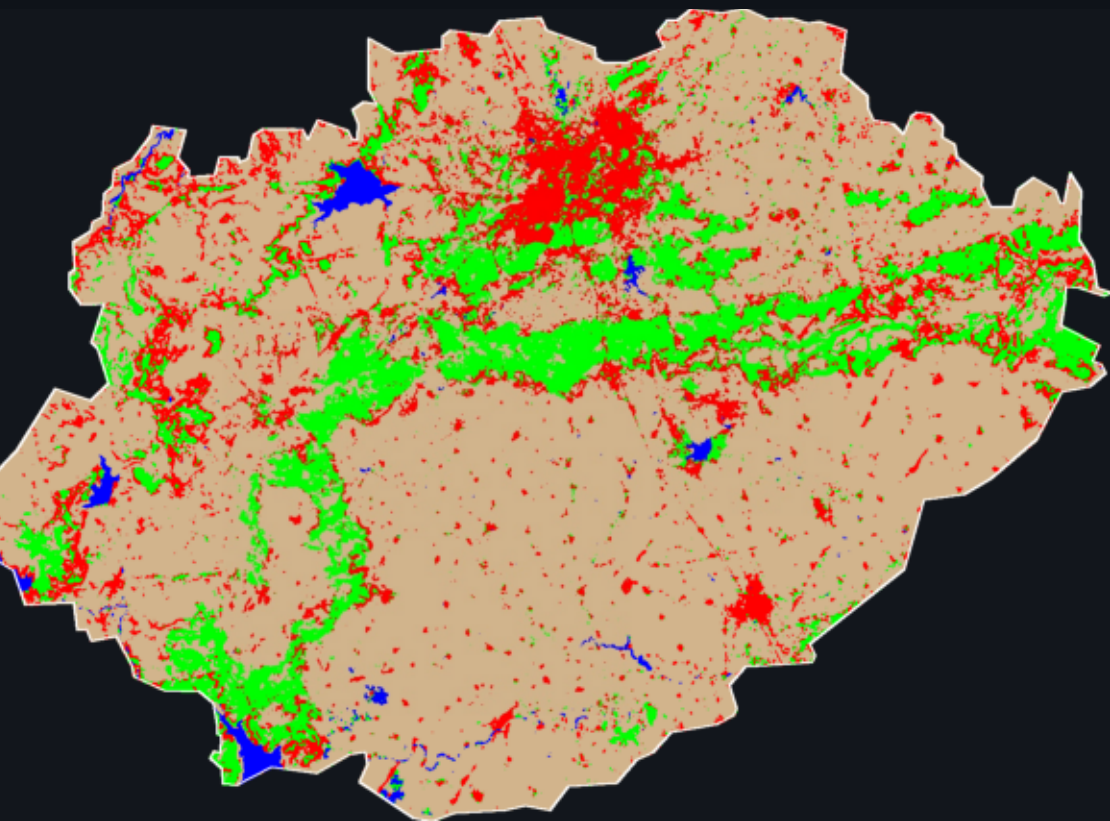
|                 |        |       |         |
|-----------------|--------|-------|---------|
|                 | before | after | change  |
| Soil recall     | 64%    | 75%   | +11.3pp |
| Buildings prec. | 62%    | 65%   | +2.9pp  |
| Soil->Buildings | 37     | 24    | -13     |
| Overall         | 65%    | 63%   | -1.8pp  |

## AREA REPORTED BY THE DASHBOARD

|           |       |            |
|-----------|-------|------------|
| Forest    | 5.0%  | 34,203 ha  |
| Water     | 3.8%  | 26,145 ha  |
| Buildings | 9.2%  | 63,128 ha  |
| Soil      | 82.1% | 566,377 ha |

# Gwalior

Best model: Random Forest · 80.2% agreement with WorldCover



Forest Water Buildings Soil

## CONFIGURATION

model Random Forest  
params 200 trees · bag 0.3 · maxNodes 10 · minLeaf 1  
composite 2025-03-01 to 2025-04-30, cloud <15%  
features 27 bands (S2 + indices + GLCM + SAR)  
training 5,000 pts (4,000 Jabalpur + 1,000 MP soil)  
smoothing 1 px focal mode

## AGREEMENT WITH ESA WORLDCOVER

| class          | recall       | precision |
|----------------|--------------|-----------|
| Forest         | 85%          | 94%       |
| Water          | 96%          | 94%       |
| Buildings      | 85%          | 63%       |
| Soil           | 55%          | 76%       |
| <b>OVERALL</b> | <b>80.2%</b> |           |

## EFFECT OF THE 1,000 NEW SOIL POINTS

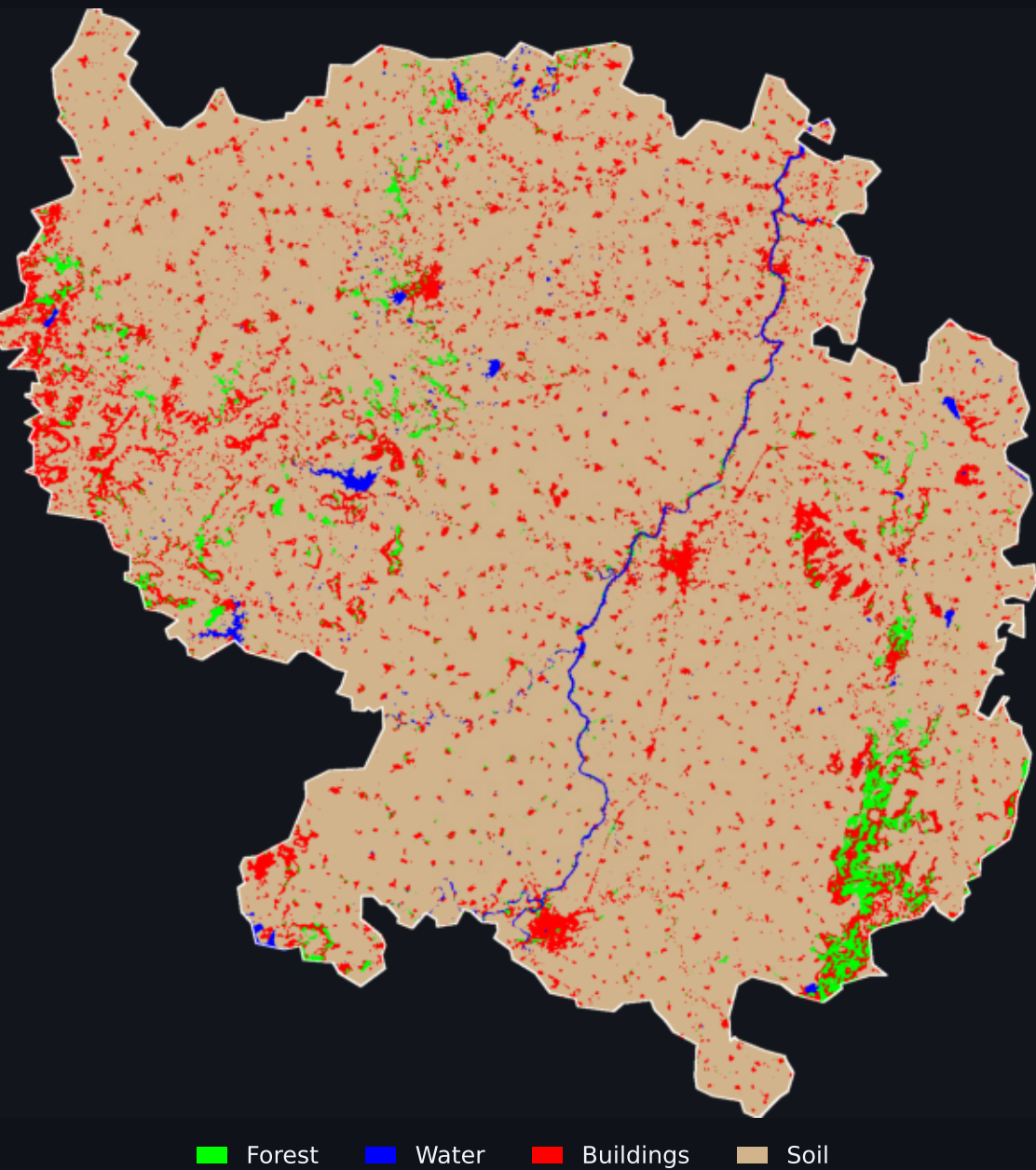
|                 | before | after | change  |
|-----------------|--------|-------|---------|
| Soil recall     | 42%    | 55%   | +12.7pp |
| Buildings prec. | 59%    | 63%   | +3.9pp  |
| Soil->Buildings | 77     | 60    | -17     |
| Overall         | 79%    | 80%   | +1.5pp  |

## AREA REPORTED BY THE DASHBOARD

|           |       |            |
|-----------|-------|------------|
| Forest    | 16.1% | 82,529 ha  |
| Water     | 1.3%  | 6,573 ha   |
| Buildings | 7.8%  | 39,863 ha  |
| Soil      | 74.8% | 382,637 ha |

# Vidisha

Best model: Random Forest · 68.8% agreement with WorldCover



## CONFIGURATION

|           |                                               |
|-----------|-----------------------------------------------|
| model     | Random Forest                                 |
| params    | 200 trees · bag 0.3 · maxNodes 10 · minLeaf 1 |
| composite | 2025-03-01 to 2025-04-30, cloud <15%          |
| features  | 27 bands (S2 + indices + GLCM + SAR)          |
| training  | 5,000 pts (4,000 Jabalpur + 1,000 MP soil)    |
| smoothing | 1 px focal mode                               |

## AGREEMENT WITH ESA WORLDCOVER

|           |        |           |
|-----------|--------|-----------|
| class     | recall | precision |
| Forest    | 51%    | 100%      |
| Water     | 81%    | 94%       |
| Buildings | 95%    | 55%       |
| Soil      | 48%    | 54%       |
| OVERALL   | 68.8%  |           |

## EFFECT OF THE 1,000 NEW SOIL POINTS

|                 |        |       |         |
|-----------------|--------|-------|---------|
|                 | before | after | change  |
| Soil recall     | 32%    | 48%   | +16.0pp |
| Buildings prec. | 51%    | 55%   | +4.1pp  |
| Soil->Buildings | 92     | 71    | -21     |
| Overall         | 68%    | 69%   | +0.3pp  |

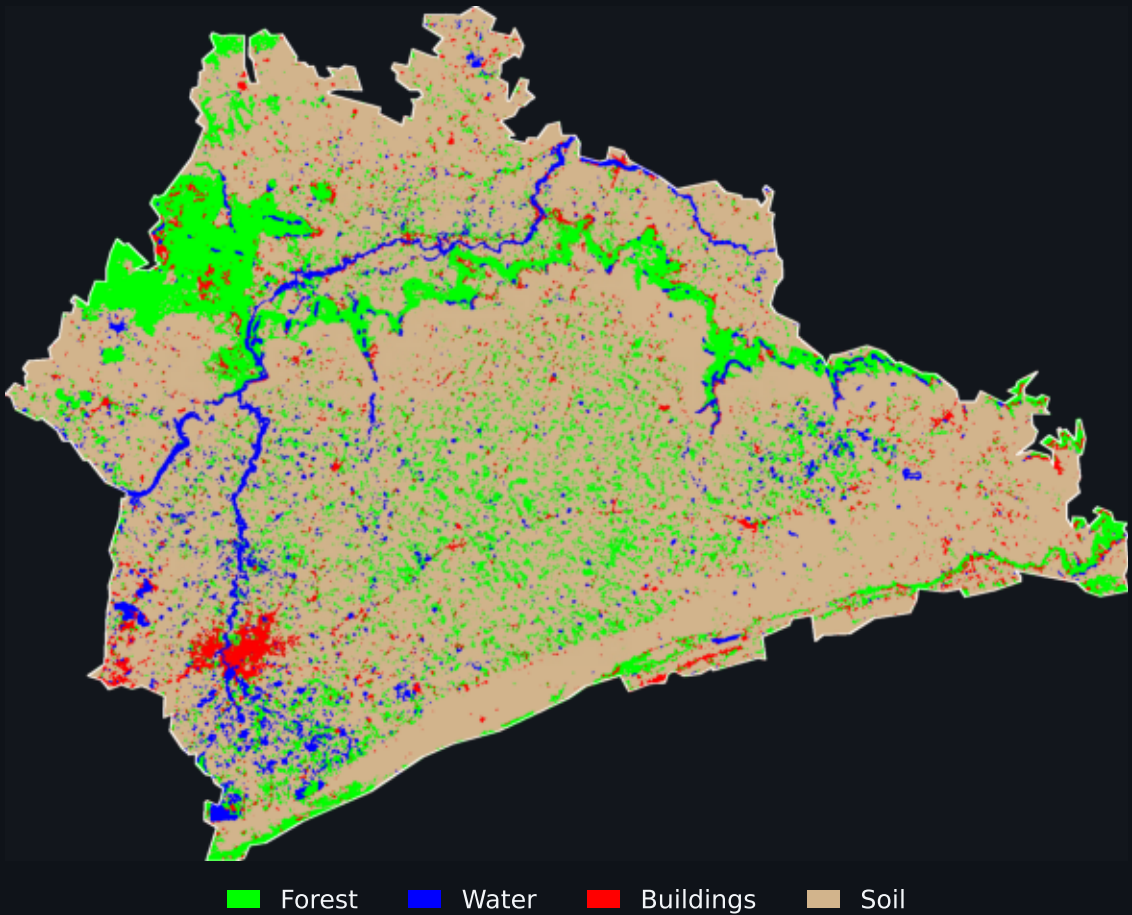
## AREA REPORTED BY THE DASHBOARD

|           |       |            |
|-----------|-------|------------|
| Forest    | 4.7%  | 37,985 ha  |
| Water     | 0.8%  | 6,574 ha   |
| Buildings | 6.1%  | 49,278 ha  |
| Soil      | 88.3% | 708,877 ha |

n = 600 stratified WorldCover points · before = 4,000 Jabalpur points, after = 5,000 including the new MP soil set

# Rewa

Best model: CART · 75.2% agreement with WorldCover



## CONFIGURATION

|           |                                            |
|-----------|--------------------------------------------|
| model     | CART                                       |
| params    | maxNodes 20                                |
| composite | 2025-03-01 to 2025-04-30, cloud <15%       |
| features  | 27 bands (S2 + indices + GLCM + SAR)       |
| training  | 5,000 pts (4,000 Jabalpur + 1,000 MP soil) |
| smoothing | 1 px focal mode                            |

## AGREEMENT WITH ESA WORLDCOVER

|           |        |           |
|-----------|--------|-----------|
| class     | recall | precision |
| Forest    | 76%    | 89%       |
| Water     | 75%    | 86%       |
| Buildings | 77%    | 72%       |
| Soil      | 73%    | 60%       |
| OVERALL   | 75.2%  |           |

## EFFECT OF THE 1,000 NEW SOIL POINTS

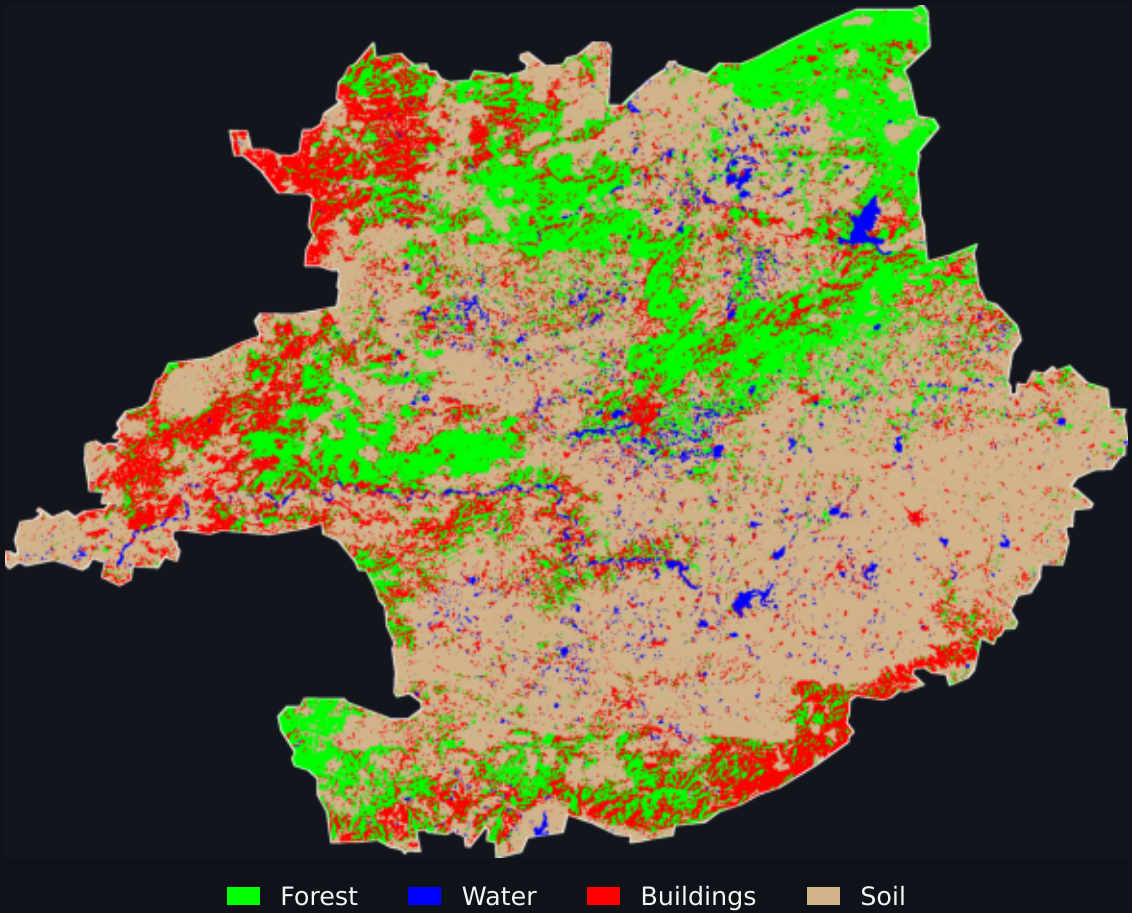
|                 |        |       |        |
|-----------------|--------|-------|--------|
|                 | before | after | change |
| Soil recall     | 67%    | 73%   | +6.0pp |
| Buildings prec. | 67%    | 72%   | +4.5pp |
| Soil->Buildings | 37     | 27    | -10    |
| Overall         | 76%    | 75%   | -0.7pp |

## AREA REPORTED BY THE DASHBOARD

|           |       |            |
|-----------|-------|------------|
| Forest    | 18.2% | 127,282 ha |
| Water     | 4.8%  | 33,705 ha  |
| Buildings | 6.1%  | 42,553 ha  |
| Soil      | 70.8% | 494,799 ha |

# Betul

Best model: CART · 64.5% agreement with WorldCover



## CONFIGURATION

|           |                                            |
|-----------|--------------------------------------------|
| model     | CART                                       |
| params    | maxNodes 20                                |
| composite | 2025-03-01 to 2025-04-30, cloud <15%       |
| features  | 27 bands (S2 + indices + GLCM + SAR)       |
| training  | 5,000 pts (4,000 Jabalpur + 1,000 MP soil) |
| smoothing | 1 px focal mode                            |

## AGREEMENT WITH ESA WORLDCOVER

|           |        |           |
|-----------|--------|-----------|
| class     | recall | precision |
| Forest    | 52%    | 96%       |
| Water     | 54%    | 74%       |
| Buildings | 84%    | 65%       |
| Soil      | 68%    | 47%       |
| OVERALL   | 64.5%  |           |

## EFFECT OF THE 1,000 NEW SOIL POINTS

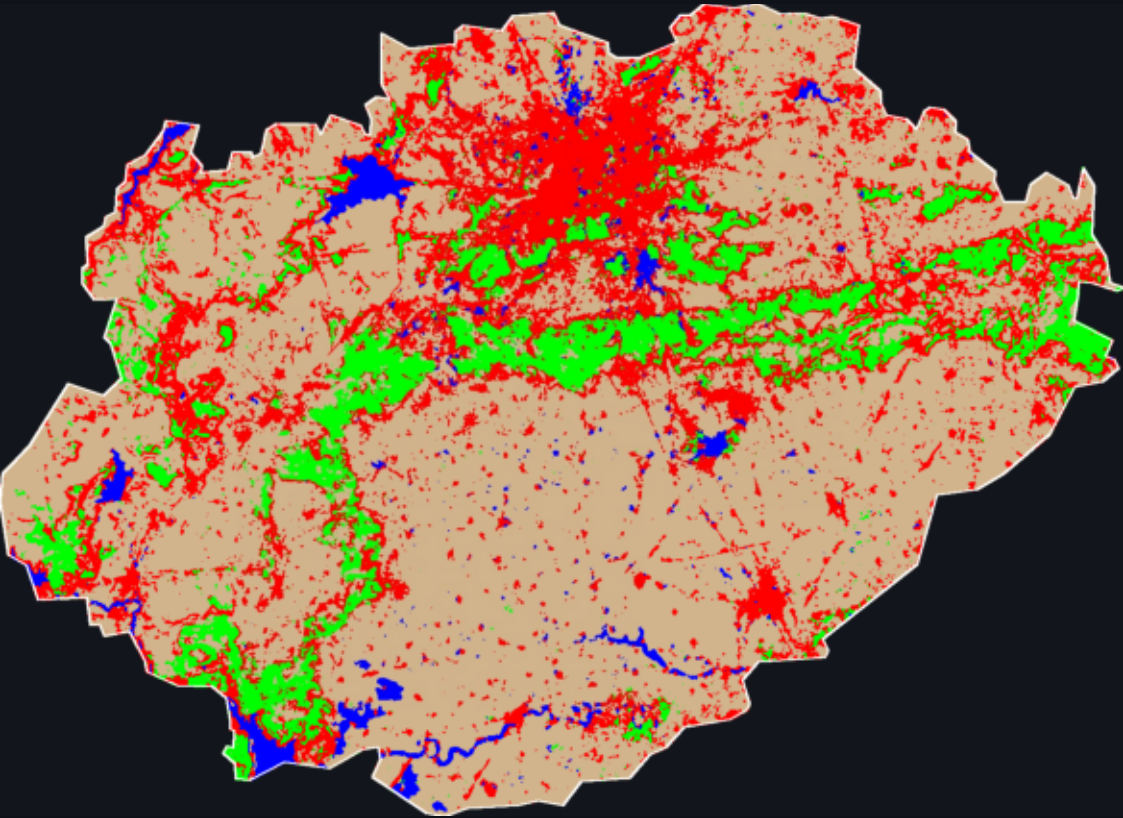
|                 |        |       |         |
|-----------------|--------|-------|---------|
|                 | before | after | change  |
| Soil recall     | 55%    | 68%   | +12.7pp |
| Buildings prec. | 69%    | 65%   | -3.7pp  |
| Soil->Buildings | 30     | 19    | -11     |
| Overall         | 69%    | 64%   | -4.3pp  |

## AREA REPORTED BY THE DASHBOARD

|           |       |            |
|-----------|-------|------------|
| Forest    | 23.8% | 259,101 ha |
| Water     | 1.3%  | 13,754 ha  |
| Buildings | 17.8% | 194,411 ha |
| Soil      | 57.1% | 622,990 ha |

# Gwalior — Gradient Boosted Trees

Gradient Boosted Trees's strongest district · 76.2% agreement



Forest Water Buildings Soil

## CONFIGURATION

|           |                                            |
|-----------|--------------------------------------------|
| model     | Gradient Boosted Trees                     |
| params    | 100 trees · shrinkage 0.1 · maxNodes 10    |
| composite | 2025-03-01 to 2025-04-30, cloud <15%       |
| features  | 27 bands (S2 + indices + GLCM + SAR)       |
| training  | 5,000 pts (4,000 Jabalpur + 1,000 MP soil) |
| smoothing | 1 px focal mode                            |

## AGREEMENT WITH ESA WORLDCOVER

|           |        |           |
|-----------|--------|-----------|
| class     | recall | precision |
| Forest    | 72%    | 98%       |
| Water     | 97%    | 87%       |
| Buildings | 96%    | 57%       |
| Soil      | 39%    | 84%       |
| OVERALL   | 76.2%  |           |

## EFFECT OF THE 1,000 NEW SOIL POINTS

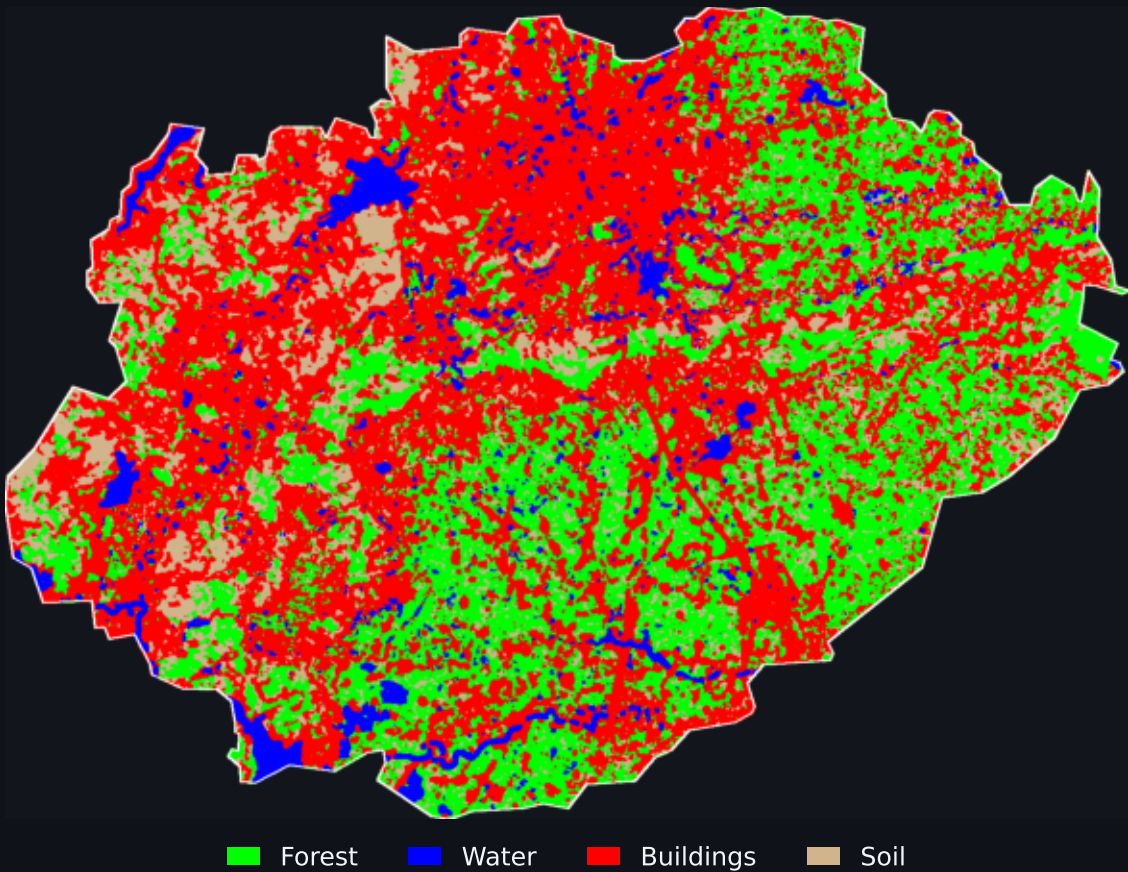
|                 |        |       |         |
|-----------------|--------|-------|---------|
|                 | before | after | change  |
| Soil recall     | 27%    | 39%   | +12.0pp |
| Buildings prec. | 53%    | 57%   | +4.1pp  |
| Soil->Buildings | 94     | 77    | -17     |
| Overall         | 73%    | 76%   | +3.5pp  |

## AREA REPORTED BY THE DASHBOARD

|           |       |            |
|-----------|-------|------------|
| Forest    | 15.4% | 78,663 ha  |
| Water     | 2.4%  | 12,160 ha  |
| Buildings | 10.6% | 54,055 ha  |
| Soil      | 71.7% | 366,725 ha |

# Gwalior — KNN

KNN's strongest district · 65.3% agreement



## CONFIGURATION

|           |                                            |
|-----------|--------------------------------------------|
| model     | KNN                                        |
| params    | k=5 · Euclidean                            |
| composite | 2025-03-01 to 2025-04-30, cloud <15%       |
| features  | 27 bands (S2 + indices + GLCM + SAR)       |
| training  | 5,000 pts (4,000 Jabalpur + 1,000 MP soil) |
| smoothing | 1 px focal mode                            |

## AGREEMENT WITH ESA WORLDCOVER

|           |        |           |
|-----------|--------|-----------|
| class     | recall | precision |
| Forest    | 50%    | 88%       |
| Water     | 98%    | 76%       |
| Buildings | 92%    | 54%       |
| Soil      | 21%    | 48%       |
| OVERALL   | 65.3%  |           |

## EFFECT OF THE 1,000 NEW SOIL POINTS

|                 |        |       |        |
|-----------------|--------|-------|--------|
|                 | before | after | change |
| Soil recall     | 15%    | 21%   | +6.0pp |
| Buildings prec. | 53%    | 54%   | +0.7pp |
| Soil->Buildings | 78     | 75    | -3     |
| Overall         | 66%    | 65%   | -0.7pp |

## AREA REPORTED BY THE DASHBOARD

|           |       |            |
|-----------|-------|------------|
| Forest    | 25.9% | 132,563 ha |
| Water     | 2.8%  | 14,100 ha  |
| Buildings | 10.7% | 54,841 ha  |
| Soil      | 60.6% | 310,099 ha |

# Model Comparison and Caveats

MODEL COMPARISON, ALL EIGHT DISTRICTS, 5,000 TRAINING POINTS

| model                  | mean overall | mean soil | wins |
|------------------------|--------------|-----------|------|
| Random Forest          | 68.0%        | 57%       | 4    |
| CART                   | 68.0%        | 67%       | 4    |
| Gradient Boosted Trees | 63.4%        | 31%       | 0    |
| KNN                    | 57.6%        | 19%       | 0    |

Random Forest and CART tie on overall agreement, but CART recovers markedly more soil and is the better choice where bare ground matters. Random Forest remains the safer default on mixed terrain.

SVM is absent. It exceeds Earth Engine's interactive compute limit on every district at 20 m, and re-running it at 60 m produced ~27% overall with 0% soil recall -- not an SVM result but a scale artefact, because GLCM and SAR texture are computed on the sampling grid and at 60 m the model sees features nothing like its 10 m training data. SVM cannot be honestly scored at district scale with this feature stack.

## WHAT THESE NUMBERS DO NOT SAY

The 1,000 new soil points lie inside these eight districts. These are therefore in-domain figures: they show the fix working where the labels were added, not generalisation to unlabelled terrain. The state-wide figure in doc/mp\_accuracy\_report.md remains the honest external number.

Soil is now 2,000 of 5,000 training points, twice every other class. Part of the recall gain is that shifted prior rather than better separation. Buildings precision rose in six districts of eight, which suggests the separation genuinely improved, but the two effects are not disentangled here.